

Data Engineering Weekly Assignments

Problem Statement

A company is introducing a new feature in its digital product and wants to determine whether the new design performs better than the existing one. Instead of releasing the feature to every user, the company conducts an **A/B Test**. One group of users (Group A) continues using the existing version, while another group (Group B) receives the new version. The objective is to determine whether the new feature improves business metrics such as user engagement, conversion rate, retention, or revenue.

To answer these questions, the organization requires a reliable data platform capable of collecting, validating, transforming, storing, and serving high-quality data for analysis.

Example

Assume the company changes the design of a button to improve user conversions. Every interaction—opening the application, viewing a page, clicking the button, completing a transaction, or exiting the application—is captured as an event. After collecting data from thousands of users, analysts compare the behaviour of Group A and Group B to determine whether the new design performs better. This analysis is only possible if a complete data pipeline exists.

Project Objective

Build an end-to-end data pipeline that enables the organization to collect, process, store, and analyze A/B testing data. The final solution should support event collection, data ingestion, warehouse modelling, pipeline automation, dashboarding, and experiment analysis.

Assignments

Assignment 1 – Business Understanding, Event Design & Data Generation

Assignment: Understand the business problem, define KPIs, design the event tracking plan and event schema, and generate realistic A/B testing event data.

Assessment (5 Marks): Business Understanding (1), Event & Schema Design (2), Realistic Data Generation (1), Documentation (1)

Assignment 2 – Data Ingestion & Data Warehouse Modelling

Assignment: Ingest the generated data into a database and build Raw, Staging, Warehouse and Data Mart layers using SQL.

Assessment (5 Marks): Database Design (2), Data Ingestion (1), SQL Transformations (1), Data Quality (1)

Assignment 3 – Pipeline Automation & Data Validation

Assignment: Automate the pipeline, implement scheduling, logging, monitoring and data validation.

Assessment (5 Marks): Pipeline Automation (1), Scheduling (1), Validation (1), Logging & Monitoring (1), Reliability (1)

Assignment 4 – Analytics Dashboard & Experiment Evaluation

Assignment: Build dashboards, calculate KPIs, perform statistical testing, and present recommendations.

Assessment (5 Marks): Dashboard (1), KPI Computation (1), Statistical Analysis (1), Business Recommendations (1), Presentation (1)

Total Continuous Assessment: 20 Marks (4 Assignments × 5 Marks)